Computer Simulation
Lesson 7: Queuing System

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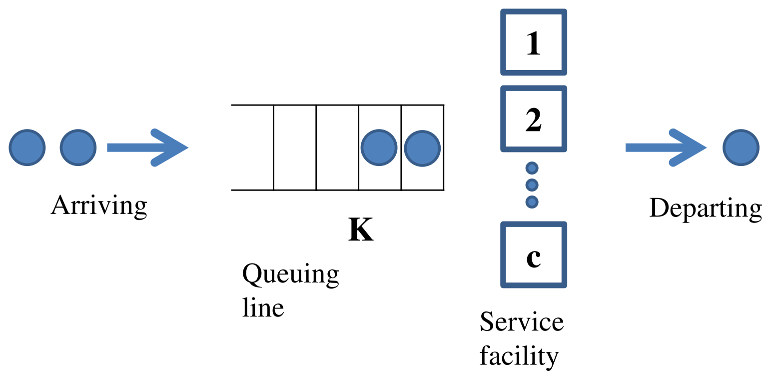
Lesson 7
Queuing System

- Introduction to Queuing System
- Kendall's Notation
- Practice
- Submission Guideline

Introduction to Queuing System

- A queuing system is a place where customers arrive according to an arrival process, obtain service from a service facility, and leave the system



Examples

- **Bus stop**: how long do students have to wait at the bus stop? Are there enough checkouts?
- **Canteen**: how many tables are necessary?
- **Mobile communication network**: users use many internet services such as downloading documents, watching video, visiting websites. How much capacity is enough? Should the data delivery for certain types of applications receive higher priority than others? How many access points need to be installed?
- **Hospital emergency department**: how many nurses and doctors need to be working to provide patients with proper care in a timely manner? How many intensive care units are required?
- **Many other situations**: ATM, bank, traffic light, telecom networks, computer systems, and so on.

Why do we need Queuing System

- To evaluate / predict system performance
  - Average number of customers
  - Average waiting time of customers
  - Measurement of busy time / idle time of servers
  - Optimizing the cost (e.g., the number of servers)
  - System utilization

  → The results are often used when making business decisions about resource needed to provide a service

How to describe the characteristics of a queuing system

- Customer arrival processes
- Service time
- Service discipline
- Service capacity
- Number of service channels

Kendall's Notation

- In queuing theory, Kendall's notation is used to characterize the parameters of a queuing system
- A/S/c/K/N/D
  - A: Probability distribution of customers' arrivals (Arrival process / inter-arrival-time distribution)
  - S: Probability distribution of service time (Service-time distribution)
  - c: Number of parallel servers
  - K: Capacity of the queue (see Wikipedia website)
  - N: Size of arriving population
  - D: Queuing discipline

A/S/c/K/N/D: Arrival Distribution

- Probability distribution arrival pattern of customers
- Commonly letters are used to describe the type of arrival distributions

Letter	Name of Distribution	Brief Description
M	Markovian (Memoryless)	Exponential distribution for inter-arrival time
E_k	Erlang distribution	Erlang distribution with shape parameter k
D	Deterministic or constant distribution	Constant inter-arrival time
G	General distribution	General probability distribution with known mean and variance

A/S/c/K/N/D : Service-time Distribution

- Probability distribution of service patterns
- Commonly letters are used to indicate the type of service distributions

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Other Notations: A/S/c/K/N/D

- c: the physical number of servers
- K: the capacity of the queue (default K = infinite)
  - The system capacity: Cs = c + K
  - The maximum number of customers that a system can accommodate (both customers at the waiting queue and customers at the service facility)
- N: the size of arriving population (default N = infinite)
  - Infinite N means that the arrival process will be unaffected by the number of clients already in the system
  - For a finite population, the customer arrival rate is a function of the number of customers in the system

A/S/c/K/N/D : Queuing Discipline

- Queuing or serving discipline refers to the way in which customers in the waiting queue are selected to serve

Symbol	Brief Description
FCFS (FIFO) (default)	First come first served or first in first out → customers are served in the order of their arrivals (normal queue operation)
LCFS (LIFO)	Last come first served or last in first out → the last customer will be served first (stack operation)
PQ (Priority)	Customers are classified according to their assigned priorities, higher priority will be served first
SIRO (Random)	Service in random order
GD	General discipline

Kendall's Notation (2)

- Simplest form of Kendall's notation: A/S/c
  →K, N, D can be omitted if K and D are assumed to be infinite, D is assumed to be FIFO
  - M/M/1: Poisson distribution for arrival and exponential distribution for service time, and one server
  - M/M/c: Poisson distribution for arrival and exponential distribution for service time, and c servers
  - M/D/c: Poisson distribution for arrival, service time is constant, and c servers
- More comprehensive model A/S/c/K/N/D is used when
  - The capacity of queuing system is finite
  - The calling source is finite

Popular Questions in a Queuing System

- When you know service demands and server capacity → find the system performance
- When you know server capacity and required performance → find acceptable demands
- When you know the service demands and required performance → find the required server capacity

Practice (queuing system simulation)

- Number of customers = 1000
- Inter-arrival time according to exponential distribution
  - 10 people per 1 minute
- Service time according to exponential distribution
  - 2 minutes per person
- **Find the optimal number of servers to let the average waiting time be less than 5 minutes**

Hint 1

- 10 people/minute → expect an average inter-arrival time of 0.1 minute and the time between individual arrivals is completely random

Hint 2

- Generate a random inter-arrival time between two customers → store in a vector
- Assume the arrival time of the first customer is 0 → then generate the arrival time of the next customer; then generate the arrival time of the next customer; …. saving to the vector of inter-arrival time
- Similarly, generate the service time of each customer and store in a vector
- To generate an exponential random number → check with exprnd()

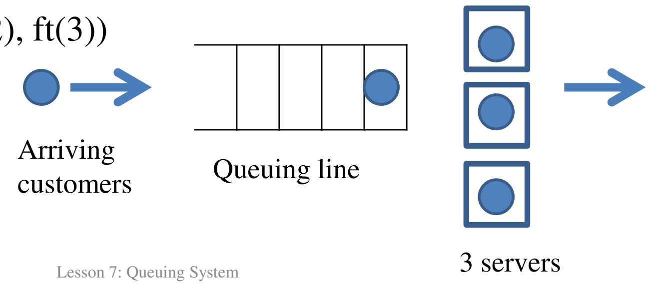
Hint 3

Assuming that there are 3 servers in the system:

1. First 3 customers will be served without waiting:
ft(i) = at(i) + st(i), i=1:3

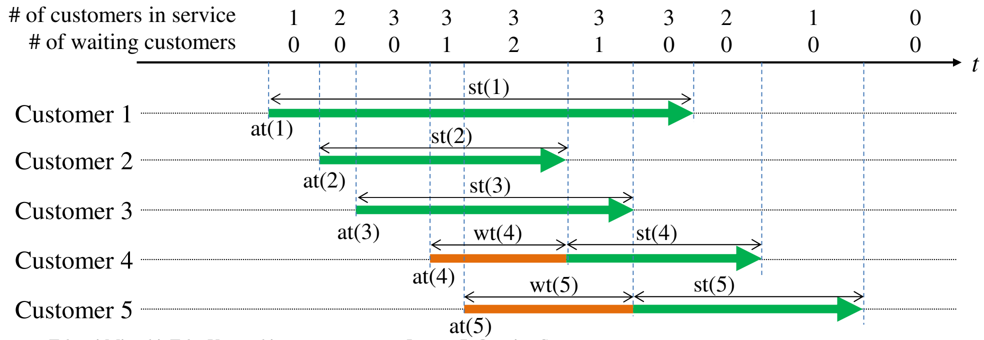
2. Fourth customer has to find which customer will leave first.
Queuing delay happens when
at(4) < min(ft(1), ft(2), ft(3))

% notation
at: arrival time
st: service time
ft: finish time



Hint 4

- Waiting time wt
- at(4) < min(ft(i)), i=1:3
- wt(4) = max(0, min(ft(i)) - at(4))



Submission Guideline

- Complete the assignment by yourself
- 11 June: Bring your solution to the class
- Divide into several teams and discuss
- 18 June: Each team will make a presentation to introduce the solution (report, MATLAB program)
- Each student will grade
  - The solutions of his / her team and other teams
  - The contribution of his / her team members

Thank you very much for your attendance!

References

- Queuing Systems

L. Kleinrock, “Queuing Systems, Volume 1: Theory,” Wiley, 1975

- Queueing Modeling Fundamentals

Ng Chee-Hock and Soong boon-Hee, “Queuing Modeling Fundamentals With Applications in Communication Networks,” 2008